

Gagan Chandra

Backend Engineer • Python • FastAPI • Flask • MongoDB • Docker • REST APIs

+91-9027319488 — gaganchandra02@gmail.com — linkedin.com/in/gagan-chandra — github.com/gagannchandra — www.gaganchandra.in

Summary

Final-year CS (AI) student at PSIT Kanpur who has spent the past two years building and shipping real software – four production apps live right now, covering REST API design, JWT authentication, ML model serving, and full-stack development. I enjoy understanding how systems work and figuring out why something breaks. Looking for a backend engineering role where I can keep learning and contribute from day one.

Technical Skills

Languages: Python, JavaScript, C++, Java, SQL, HTML5, CSS3

Backend & APIs: FastAPI, Flask, Node.js, Express.js, RESTful APIs, REST API Design, JWT Authentication, Role-Based Authorization, Pydantic, SQLAlchemy, express-validator

ML / AI: Scikit-learn, Pandas, NumPy, XGBoost, Ensemble Learning, Feature Engineering, Model Deployment, LLM API Integration

Databases: PostgreSQL, MongoDB, MySQL, SQLite, Redis, PyMongo, Mongoose

Frontend: React 19, Vite, Tailwind CSS, Recharts, Vanilla JS

DevOps & Tools: Docker, GitHub Actions, Git, pytest, unit testing, Agile, Postman, Linux, Render, Railway, Vercel

Experience

Open Source Contributor

Python, JavaScript

Remote

Jan 2026 – Present

- Contributing bug fixes, test coverage improvements, and documentation to open-source Python and JavaScript projects; working with maintainers through pull requests, code reviews, and issue triage.

Projects

HealthAI: Disease Prediction System · Python, FastAPI, Scikit-learn, XGBoost, React 19, Docker · Live · GitHub

- Trained a soft-voting ensemble of Decision Tree, Random Forest, and Naive Bayes on 102,238 samples across 141 disease classes and 343 features, reaching **87.84% accuracy**; soft-voting produces more stable predictions than any single classifier. Served via FastAPI + Pydantic v2 at 5.3 ms avg response time and 187 req/s per core.
- Built a React 19 + Vite frontend with voice symptom input (Web Speech API) and Recharts confidence charts. Each prediction returns the top-3 diagnoses with specialist recommendation, medications, diet, and precautions sourced from structured medical CSVs and a pre-built JSON knowledge base.

GitHub Repository Health Checker · Python, FastAPI, GitHub REST API, LLaMA 3.1 · Live · GitHub

- Fetches 7 repository metrics concurrently from the GitHub REST API, formats them as structured context, and sends to NVIDIA-hosted LLaMA 3.1 8B Instruct – response is parsed into strengths, concerns, a health summary, and a score out of 100.
- A GitHub Personal Access Token raises the rate limit from 60 to 5,000 req/hr, making concurrent fetching across large repositories reliable. Status badges (Active / Inactive / Needs Review) assigned deterministically from activity signals; deployed as an async FastAPI service on Render.

Shortly: URL Shortener with Analytics · Python, Flask, MongoDB, Docker, GitHub Actions · Live · GitHub

- Started as a weekend Flask + SQLite project. Rebuilt in v2: swapped to MongoDB with TTL indexes so links expire automatically, added QR code generation, per-IP rate limiting (Flask-Limiter), security headers (Flask-Talisman), and a multi-stage non-root Docker image. Deployed on Render via Gunicorn.
- Refactored to an app-factory pattern with Blueprint routing and a service layer – mainly to make the codebase testable. Added a pytest suite using mongomock and a GitHub Actions CI pipeline that runs tests and builds the Docker image on every push to main.

TaskFlow: Team Task Manager · React 19, Node.js, Express.js, MongoDB Atlas, JWT, Railway · Live · GitHub

- MERN stack task manager with JWT auth, bcrypt hashing, and role-based middleware – Admins get full CRUD; Members can view and update only their assigned work. Mongoose models (User, Project, ProjectMember, Task) with express-validator on all requests; Kanban board, analytics dashboard, and deployed as a monorepo on Railway where Express serves the React build directly.

Education

Pranveer Singh Institute of Technology (PSIT), AKTU

Kanpur, Uttar Pradesh

B.Tech – Computer Science and Engineering (Artificial Intelligence Specialization)

July 2026

- Coursework:** Data Structures & Algorithms, OOP, Operating Systems, DBMS, Machine Learning, Computer Networks, Software Engineering

Achievements & Certifications

GATE 2026 – Qualified in both CS & IT and DS & AI streams

520+ DSA problems solved across LeetCode, GeeksforGeeks, and HackerRank; 5-Star rating in Problem Solving and Python on HackerRank

Salesforce Agentblazer Champion – Mountaineer Rank (46+ badges)

CS50: Introduction to Computer Science (Harvard) — Google AI Essentials — Introduction to Data Science, IBM (edX) — Agile with Atlassian Jira (Coursera)